

BLV Series Motor Driver – Interface Reference

Quick guide for direct CN4 control with Arduino Mega. For BLV620KM, BLV640KM and similar 24 V drivers.

1. Connectors

Connector	Purpose
CN1	DC power supply input. Connect 24 V battery (+) and (-).
CN2	3-phase output to motor. The driver sends PWM voltage here.
CN3	Motor sensor signal. Connect motor sensor cable.
CN4	I/O signal port. D-Sub 15-pin. Used by Arduino to control the driver.
CN5 / CN6	RS-485 communication ports. RJ-45. Used for Modbus control
CN7	Service port for OPX-2A or PC software (MEXE02). Mini-DIN 8-pin.
CN8	Electromagnetic brake wires. Connect to brake-type motor only.

2. CN4 I/O pins

CN4 Pin	Signal	Function	Logic in code (sink mode)
1	FWD	Forward run command	LOW = motor runs forward
2	REV	Reverse run command	LOW = motor runs reverse
3	STOP-MODE	How the motor stops	LOW = decel stop, HIGH = instant stop
4	M0	Operation data select	HIGH = uses data No.0 (default)
5	IN-COM	Common ground for inputs	Connect to Arduino GND
10	MB-FREE	Manual brake release	HIGH = brake auto operates
12	VM (analog)	Speed setpoint input	0 to 5 V DC = 0 to max RPM

Sink logic note: Arduino pin LOW = signal ON. Arduino pin HIGH = signal OFF. This is set by SW1-No.3 = OFF.

3. LED status

LED	Color	Meaning
POWER	Green	Lit when CN1 power is OK.
ALARM	Red	Blinks when an error is active. Count blinks to read the code.
C-DAT	Green	Flashes during RS-485 data exchange (Modbus mode only).
C-ERR	Red	Lit/blinking when RS-485 communication error happens.

4. Switch and potentiometer settings (working driver)

Switch/Pot	Setting	Meaning
SW1 (1, 2, 3)	All OFF	Sink logic, slow response, 0–5 V analog input range.
SW2-3, SW2-5	ON	RS-485 baud 115,200 bps and Modbus protocol enabled. Not used by CN4 control.
SW3	1	Slave address (only used for RS-485).
VR1 (Speed)	Free	Internal speed pot. Not used when M0=HIGH and external VM is active. Very important if used the
VR2 (Accel)	Fully CCW	Acceleration / deceleration time. Fully CCW = 0.2 sec (fastest).
VR3 (Torque)	Fully CW	Torque limit. Fully CW = 200% (maximum). Do not lower.

5. Speed control method

The Arduino sends PWM (0 to 255) on pin 2 (left) or pin 3 (right). The PWM signal becomes a DC voltage from 0 V to 5 V at the VM input on CN4. The motor runs only when FWD or REV is also ON. Speed is set by PWM value: 0 = stop, 255 = max RPM.

6. Tested working program

The test code drives two BLV drivers (left and right motor) using the pins shown in section 2. The motors run forward for 3 seconds, then stop for 3 seconds, in a loop. M0 is set HIGH so the driver uses operation data No.0 with the external analog input (VM) as the speed source. MB-FREE is set HIGH so the brake operates automatically (locks when stopped, releases during run).